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10+ Years of 3D Printing Excellence



FRACKTAL APOLLO 350 SLS 3D Printer



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HEAVY INDUSTRIES

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Foundation for Science
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Suitable for a wide range of applications

Ideal for Low - Mid Volume Production



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FRACKTAL
WORKS

Apollo 350 SLS

India's most advanced SLS 3D Printer

Selective Laser Sintering (SLS) is a transformative technology in 3D printing, enabling the creation of complex, durable parts with precision. It offers unmatched flexibility and efficiency, revolutionizing manufacturing.

The Apollo 350 SLS developed under Phase-II of the Ministry of Heavy Industries' Capital Goods Scheme in collaboration with IISc and the Foundation for Science Innovation and Development, represents a significant milestone in India's advanced manufacturing journey. This polymer 3D printer combines cutting-edge technology with user-focused innovation, making high-performance additive manufacturing accessible and efficient.

Key features of the Apollo 350 SLS include a highly stable CO2 laser, a precision scanning head, and an advanced preheating system that minimizes powder degradation and maximizes reusability. Its removable build module, which houses the overflow tank, Z-axis, and feed axis, allows for continuous operation—boosting throughput and productivity.

Designed to serve a wide range of industries, from service bureaus to mass production, the Apollo 350 is the foundation for future advancements in polymer 3D printing, paving the way for machines with larger bed sizes, advanced energy sources, and multi-laser configurations.

The Apollo 350 isn't just a 3D printer—it's a step toward transforming India's manufacturing landscape, making additive manufacturing faster, more sustainable, and more accessible than ever.

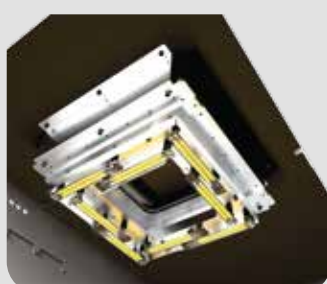


Key Features of Apollo 350 SLS



Air Curtains

User serviceable Optics Windows and Patented Air Curtain design for 24x7 production capability



LWIR heaters

8 Channel LWIR heaters for even and efficient heating across build plane.



IR Camera + RGB Camera

IR Thermal Camera + RGB Camera for 8 Zones of Process Temperature monitoring and in-situ error detection



Removable Z Axis

Removable Z Axis for Fast material and build changeover for low machine downtime and 24x7 productivity



DETAILED SPECIFICATIONS

FRACKTAL WORKS APOLLO 350 SLS 3D PRINTER

GENERAL

Product Type	Industrial Production SLS Printer
Applications	Medical, Aerospace, Industrial, Automotive, Ducting, Consumer goods
Duty Cycle	Daily production, High volume productivity

PRINTER HARDWARE

Laser Power Type	120 W CO ₂ Laser – Industrial grade with 3 Axis Galvo scanners with Z-coil
Build Volume (Max)	350 x 350 x 500 mm (XYZ)
Volume Build Rate	3 l/hr
Recoater	Variable speed Counter Rotating Roller
Powder Management	Removable hopper and Z Axis for Raw Material and Depowdering Transport and quick material change at Material Station and Depowdering Station.
Powder Recycling	% blend is material dependent and performed by Material Station
Thermal Management	Closed loop thermal control with IR camera with 8 discrete heating zones.
Build Turnover Time	Approximately 4 hours to remove from printer; new build may start; 10 hours to part in hand

PRINTER SETUP

Infrastructure	Requires compressed air, nitrogen, three-phase power, and large door openings for installation
Full Solution Footprint	Complete Solution: Printer, Material Station, Depowdering Station, N2 Generator, Vacuum – 18.8 m ² / 201 sq ft
Work Environment	Regular maintenance and floor cleaning required

PRINTER OPERATION

Software	Autodesk Netfabb, Printer UI
Operation	Mouse, keyboard, screen, and a window for surveillance
Build Monitoring	Performed at the printer or desk PC
Part Packing Density	Part spacing 5 mm; more freedom for packing in Z – geometry dependent

MATERIALS & CAPABILITY

Materials	6+ materials: PA11, PA12, GF, AF, CF, HST; pure white, black, aluminum metallic, more
Accuracy	249 µm + 100 µm each 25 mm; printer tuning enables enhanced capability
Surface Finish	Industry-leading part quality with consistent smooth parts without orange peel
Mechanical Properties	Part density, tensile, flexural, and impact properties are similar

Robust Thermoplastics for a Variety of Applications

Produce tough, durable parts from the wide materials portfolio that has been optimized, validated and tested to ensure quality, with uniform 3D mechanical properties. When you compare material properties, you'll find SLS materials compare very well with common injection molding materials. These materials are ideal for both production and prototype parts.

Applications

Housings

Manufacture in small to medium lot sizes, and bridge the time until final tools are manufactured.

Jigs and Fixtures

Print complex assembly aids and free up CNC time for other projects.

Machinery Components

Integrate functionality and replace complex assemblies.

Medical Devices

Production of patient-specific medical devices

Ducting

Optimize flow and fit in tight spaces with the freedom to print duct-work that is impossible to mold.

Consumer Goods

High-speed production for small lots and custom products.

Functional Testing

Test your prototypes for functionality such as heat run cycle tests

How SLS 3D Printing works?

Our Selective Laser Sintering (SLS) 3D printers use advanced technology to produce durable and complex parts with precision. Here's how the process works:

1. Design the Part

Start with a digital 3D model, created using CAD software. The design is prepped for printing using specialized slicing software.

2. Prepare the Powder

The printer uses fine, powdered material (commonly nylon or other engineering-grade polymers) as the raw material.

3. Layering the Powder

Layering the Powder
A thin layer of powder is spread evenly across the build platform.

4. Laser Sintering

A powerful laser precisely scans the layer, heating and fusing the powder particles only in the areas required for the part. The rest of the powder remains loose, acting as support for the structure.

5. Building Layer by Layer

The process repeats, with new layers of powder spread and selectively sintered, until the part is fully built.

6. Cooling and Finishing

After printing, the part is allowed to cool before being removed from the loose powder. The excess powder is recycled, and the part may undergo post-processing, like cleaning or surface finishing.

About Fracktal Works

Fracktal Works was founded in 2013 by two engineering students of Manipal Institute of Technology. They worked on building robots with the university robotics team, Robomanipal and eventually founded Fracktal Works.

Our mission is to empower the sustainable development of humankind with efficient problem solving that leverages the disruptive advantages of Digital Fabrication & 3D Printing.

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